

T-97700 LARGE-PRIME PARITY BOUNDARY

A no-go theorem for LAPBR67 and an exact signed Type-II frontier
 Agentic Polymath Project - 18 August 2026

STATUS

The adaptive small-prime cube is retained. The proposed nonnegative large-prime residual LAPBR67 is disproved. The replacement is an exact largest-prime Bellman identity and the open one-sided signed Type-II estimate BLPTE67. RH is unproved.

1. ADAPTIVE-DEPTH NO-GO

The published depth satisfies $L=O(\log \log \log X)$, while the total reciprocal mass of active rough primes is asymptotic to $\log \log X$. A product-safe lower bound for the final odd elementary-symmetric layer shows that it dominates all preceding layers. Consequently the full depth- L current is eventually negative, and the large-prime residual is more negative than the positive small-prime cube. Therefore LAPBR67 is false; no constant adjustment repairs it.

2. COMPLETE SMALL-PRIME CUBE

Let $Z=(\log X)^{1/4}$ and Q_Z be the product of primes $67 \leq p \leq Z$. Define

$$\begin{aligned} B_Z(X) &= \sum_{d|Q_Z} \mu(d) b(X/d)/\sqrt{d}, \\ U_Z(X) &= B_Z(X)/\sqrt{X}. \end{aligned}$$

Using $b(Y)=a_* \sqrt{Y}+c_*+O(Y^{-3/2})$ uniformly over $d|Q_Z$ gives

$$\begin{aligned} B_Z(X) &= a_* \sqrt{X} \prod_{67 \leq p \leq Z} (1-1/p) \\ &\quad + c_* \prod_{67 \leq p \leq Z} (1-1/\sqrt{p}) + o(1), \end{aligned}$$

so $B_Z(X) > 0$ for all sufficiently large X and $U_Z(X)$ is comparable to $1/\log Z$.

3. LARGEST-PRIME BELLMAN OWNERSHIP

Adjoin the primes $p > Z$ in increasing order. The exact normalized recurrence is

$$U_i(Y) = U_{(i-1)}(Y) - p_i^{-1} U_{(i-1)}(Y/p_i).$$

Telescoping gives

$$U_{\text{full}}(X) = U_Z(X) - \sum_{Z < p \leq X/2} p^{-1} U_{<p(X/p)}.$$

Every nonempty large-prime history is owned exactly once by its largest prime. Expanding the child yields the source-faithful signed correlation

$$\begin{aligned} U_{<p}(Y) &= \sum_{\substack{v \text{ squarefree, } Z < P^-(v), \\ \mu(v) U_Z(Y/v)/v.}} P^+(v) < p \end{aligned}$$

4. TYPE-I / TYPE-II SPLIT

Split at $p=X/Z$. For $p > X/Z$, the child endpoint is below Z , so no additional large prime can be active. This terminal Type-I range satisfies

$$I_Z(X) = O((\log Z)^2/\log X) = o(U_Z(X)).$$

The remaining signed term is

$$\begin{aligned} T_Z(X) &= \sum_{Z < p \leq X/Z} p^{-1} \\ &\quad \sum_{\substack{v \text{ squarefree, } Z < P^-(v), \\ \mu(v) U_Z(X/(pv))/v.}} P^+(v) < p \end{aligned}$$

BLPTE67 is the one-sided estimate

$$T_Z(X) \leq U_Z(X) - I_Z(X).$$

It is source-faithful and preserves every Mobius sign and largest-prime owner. No unrestricted reservoir or source-blind large sieve is used.

5. EXACT BOUNDARY

The Bellman identity gives

$$U_{\text{full}}(X) = U_Z(X) - I_Z(X) - T_Z(X).$$

Therefore BLPTE67 implies $U_{\text{full}}(X) \geq 0$. The inherited annular Mellin-Landau consumer would then imply RH. T-97700 does not prove BLPTE67 or RH.

THEOREM LEDGER

Adaptive small-prime cube	RETAINED POSITIVE
LAPBR67	REFUTED
Complete small-prime cube	PROVED POSITIVE
Largest-prime Bellman identity	PROVED EXACT
Terminal Type-I	CLOSED
BLPTE67	OPEN / RH-BEARING
Riemann Hypothesis	UNPROVED

REPLAY VERDICT

PASS_T97700_LAPBR_NO_GO_AND_TYPEII_REDUCTION_ALGEBRA

Proof object:

d01eaca56c2277f7244c29b57ef2902d70b2ef157bad02d8afe01ca26dc4310d